

Efficient Programmer

Problem Description

You are a programmer who has n days to arrange your work schedule. The company provides you with a task list $nums$, where:

- $nums[i]$ means: after finishing work on day i , the next time you work can be postponed at most until day $i + nums[i]$.

Therefore, if you work on day i , your next working day can be day $(i + j)$

- $1 \leq j \leq nums[i]$

For example:

If $nums[i] = 4$, it means:

- You must work on day i
- You may rest or work on days $i + 1$, $i + 2$, and $i + 3$
- The latest day you can schedule your next work is day $i + 4$

Your goal:

- You must start working from day 0
 - You must eventually complete the work on day $n - 1$
- But you want to minimize the number of working days.

Please compute:

In order to finish the task on day $n - 1$, what is the minimum number of days you need to work?

The problem guarantees that there exists at least one valid schedule to complete the work on the last day.

Input and Output

Input Format

The first line contains an integer n , representing the length of the array.

The second line contains n positive integers representing the task list.

Output Format

Output a single integer representing the minimum number of working days required to complete the task on day $n - 1$.

Constraints

- $1 \leq n \leq 100$
- $1 \leq \text{nums}[i] \leq 99$

Sample Data

Sample Input 1:

```
5
2 3 1 1 3
```

Sample Output 1:

```
3
```

Sample Input 2:

```
5
4 3 2 1 2
```

Sample Output 2:

```
2
```